



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 18: Path planning

Marius Juston

DATE

Talk to the person next to you and answer these questions.

- Last lecture we said we used “BFS”, what does it stand for? What type of data structure is this most commonly used?
- Open on the lecture’s website the path planning visualization that was made for this lecture (https://marius-juston.github.io/SE-423---Class-Material/path_planning.html), play around and draw some things and press the run button. You don’t need to understand what A* Search, BFS, Dijstras is yet, but I would recommend having this up during lecture so that you can see the different algorithms and settings

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Starting Lab 6 this week. This a 3-week lab!

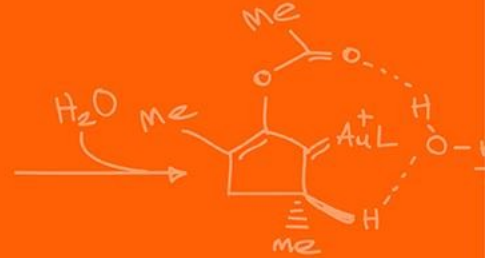
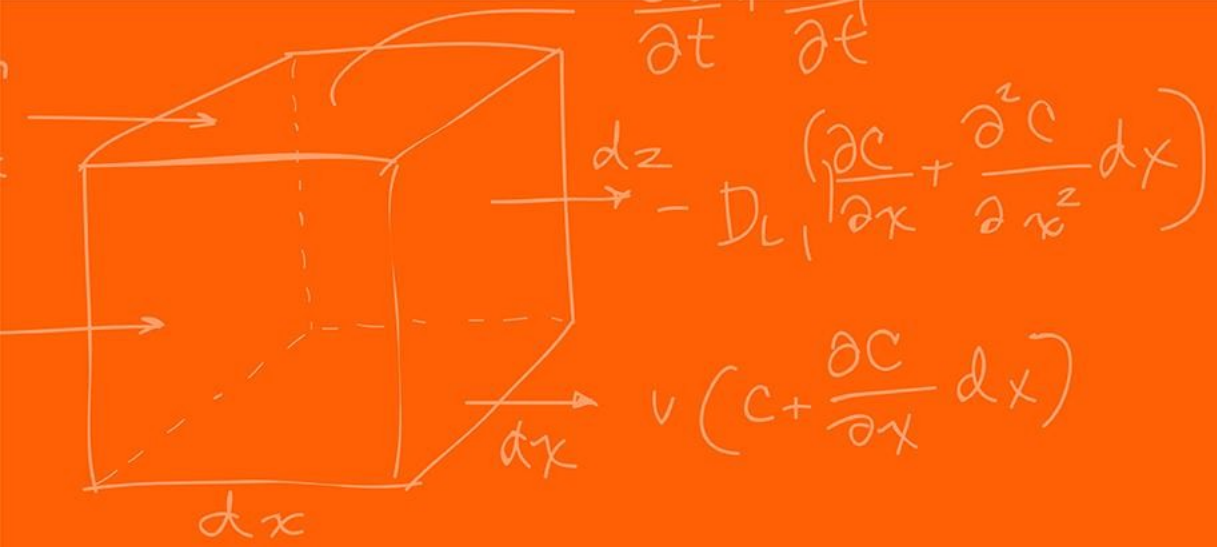
- Don't forget to have the Lab LabVIEW application implemented before lab.

Homeworks:

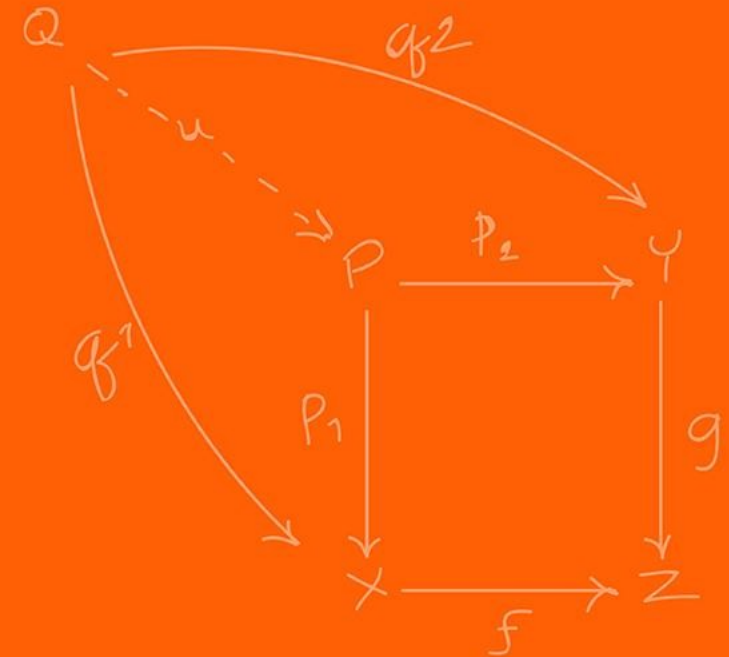
- Check-offs for [homework 4](#) Ex 2 and 3 are due **March 31, 5PM (The end of office hours)**
- Check-offs for [homework 4](#) Ex 4 are due **April 7, 5PM (The end of office hours)**
- Remainder of [homework 4](#) due on Gradescope on **April 8, 9AM**

Questions?

Homework, Lab, LabVIEW?



Path planning



I would HIGHLY recommend you looking at:

- <https://theory.stanford.edu/~amitp/GameProgramming/MapRepresentations.html>
- <https://www.redblobgames.com/pathfinding/a-star/introduction.html>

This is a series of interactable tutorial for path planning, it is very useful and goes through all the material in this lecture!

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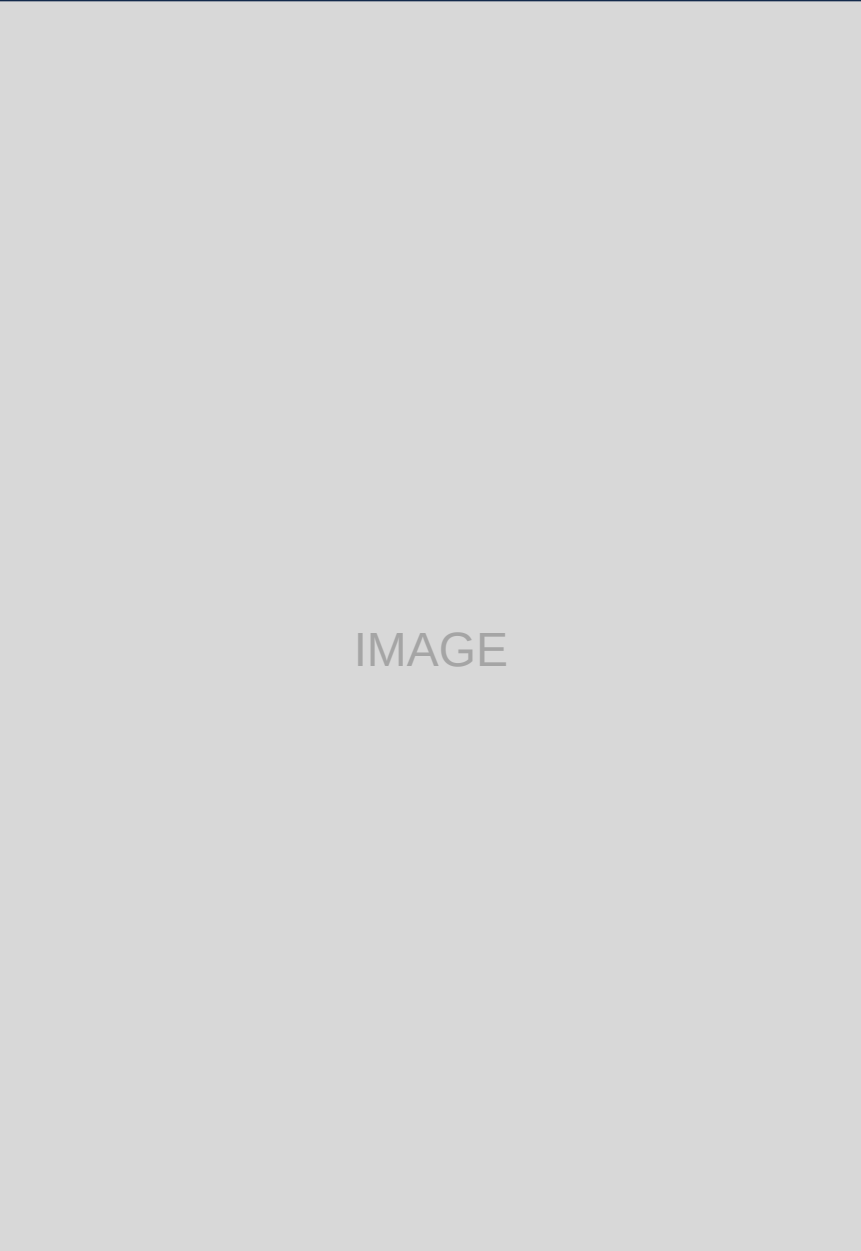


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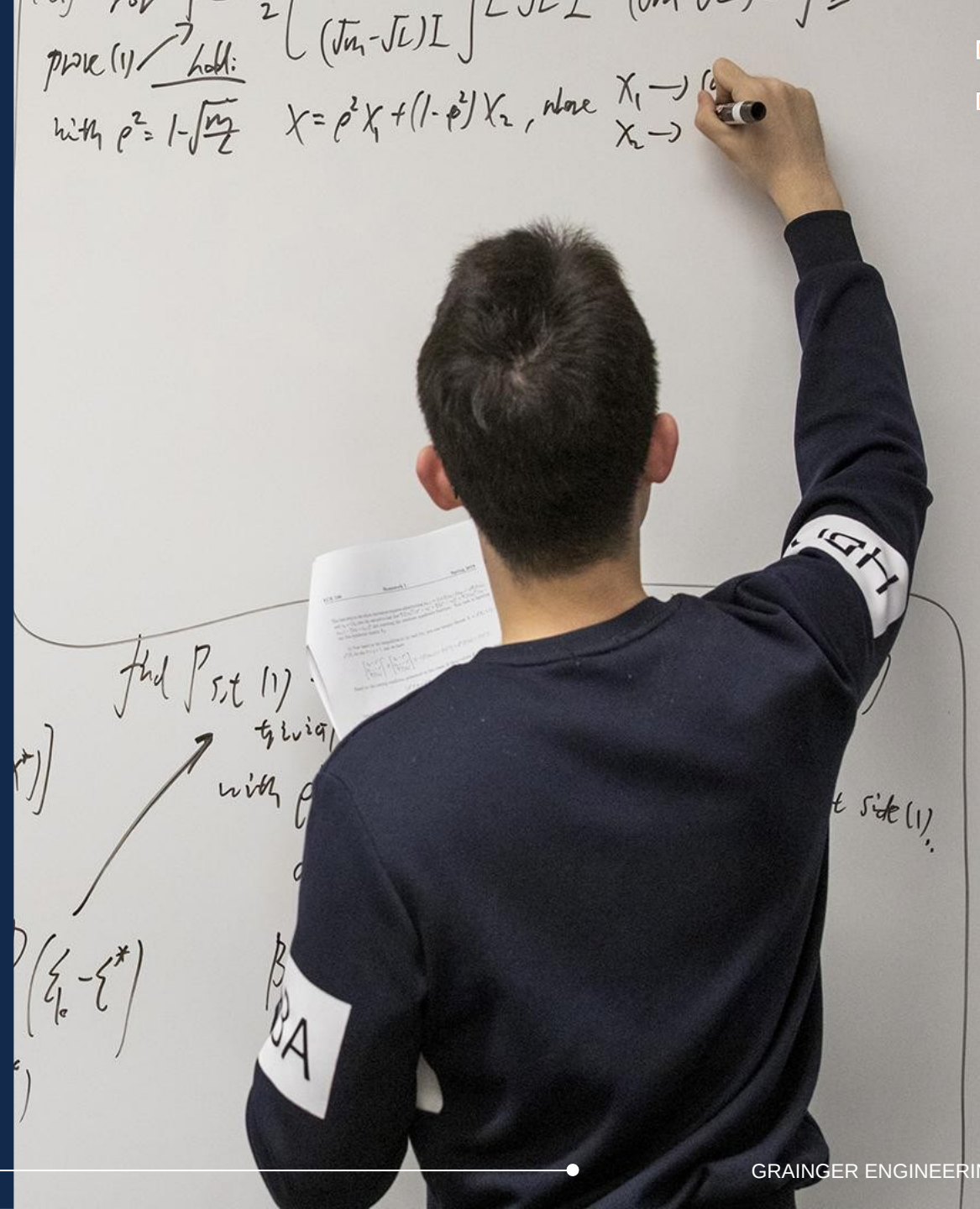
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IMAGE

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Summary

Thank You

Questions

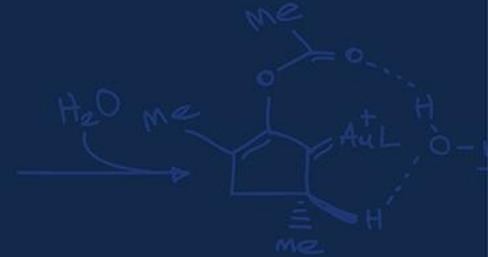
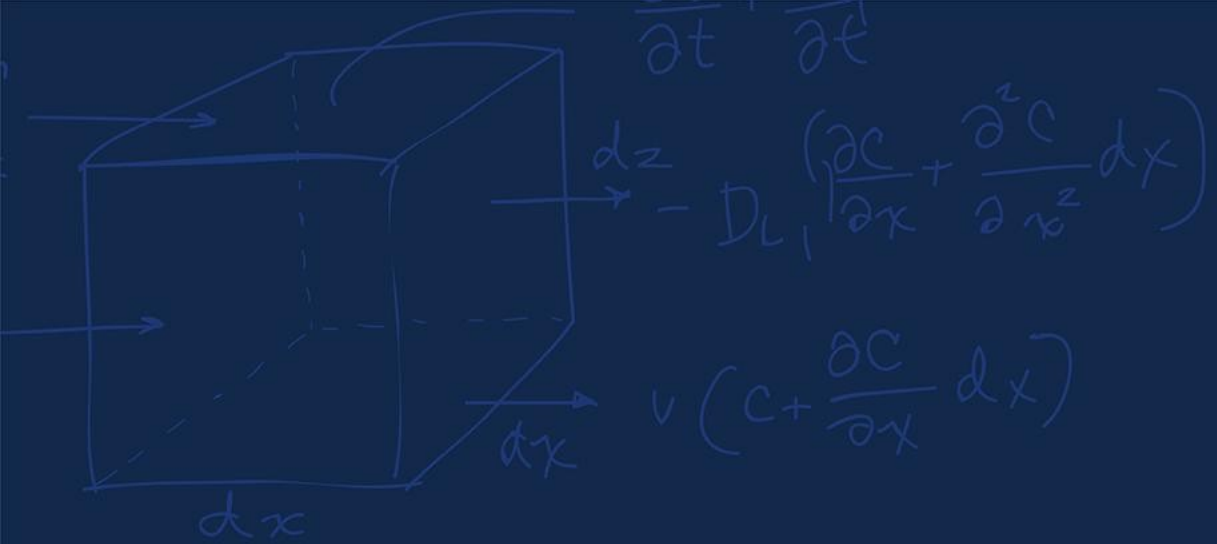
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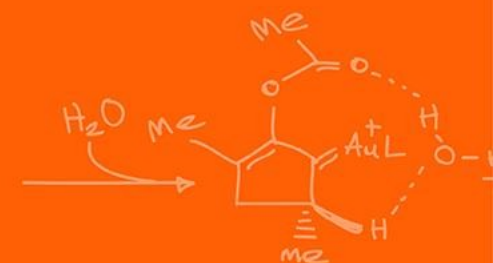
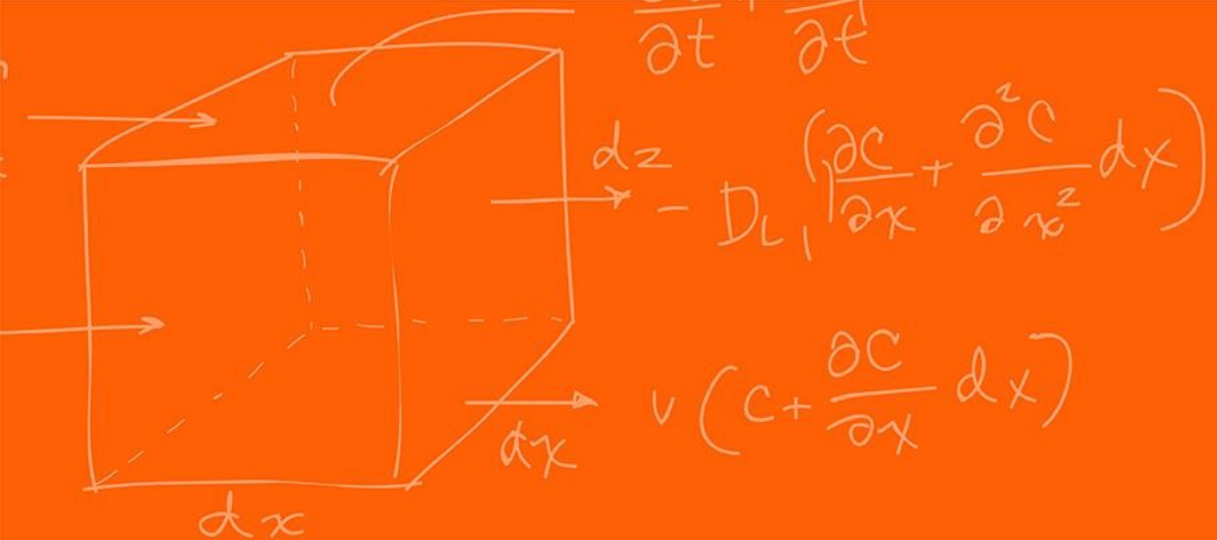
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Questions?





The Grainger College of Engineering

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